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Subject - DBMS

Que- What are key reasons why a DBMS is a preferable to a traditional file system? How does it ensure data security & consistency?

Que- A Hospital wants to store patients doctor, and appointments data in a DBMS.
Identify entities, attribute & relationships in systems. Draw ER diagram & explain key design decisions you made.

Que- An online shopping platform needs to store data about customers, order & products:
Suggest primary keys, foreign keys & integrity constraints for system. How do these constrains help maintain accuracy of Data.

Que- Why is understding ER models important when designing a database? What could happen if a relationship is not correctly defined between entities?

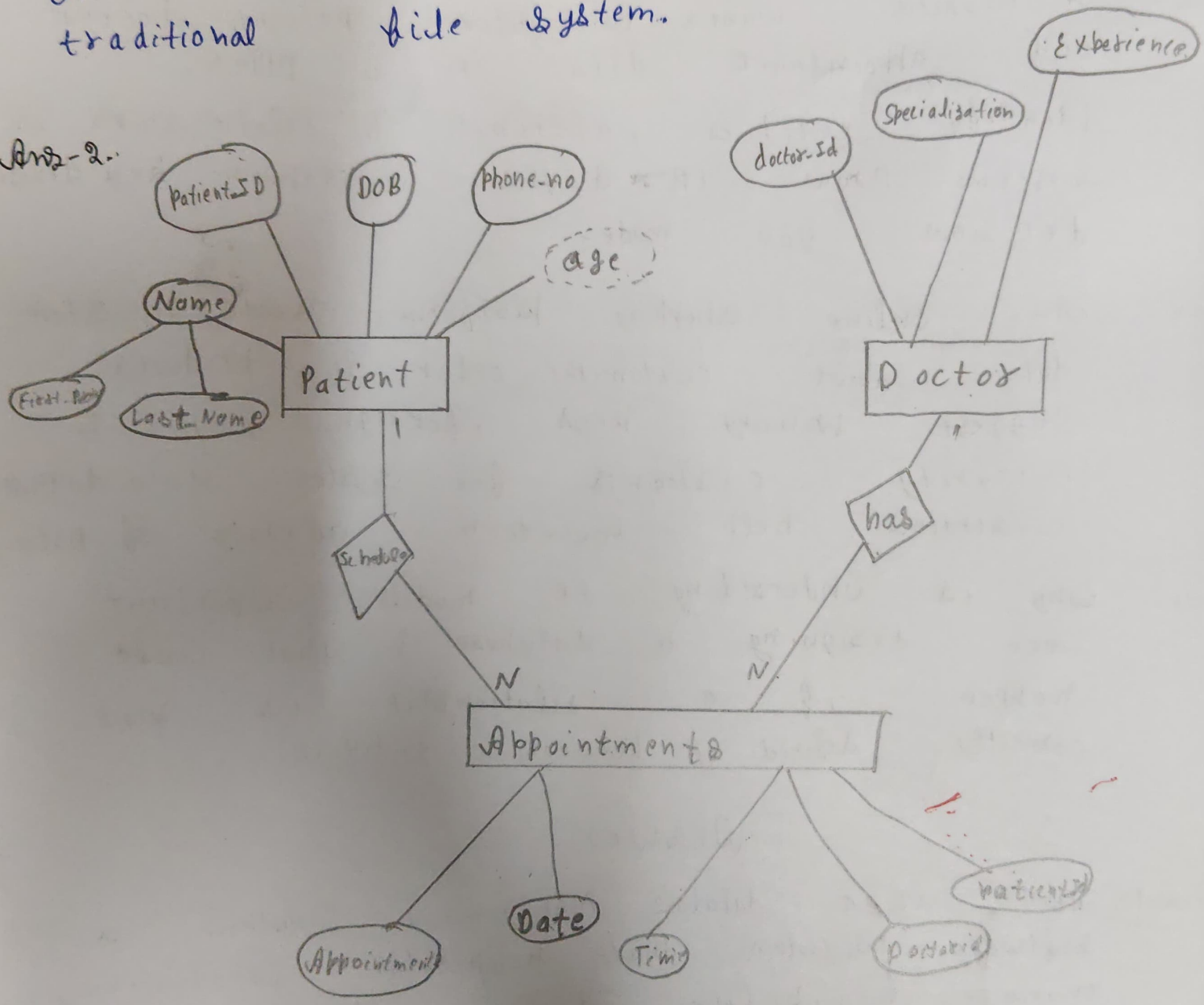
Points

Answer

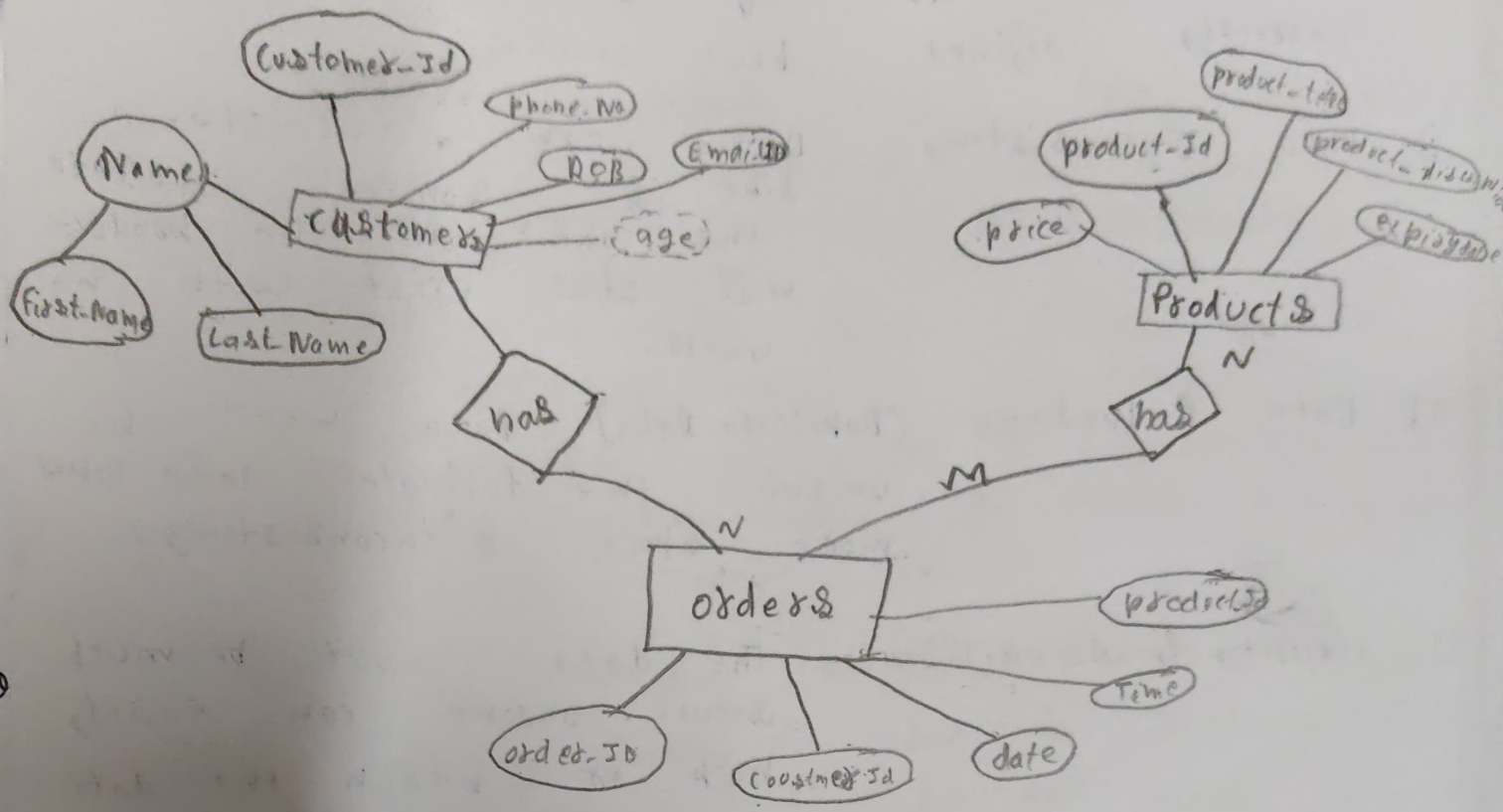
Ans-1. DBMS is a database Management System a software system that help data create, ~~image~~, manage, manipulate data efficiently.
whereas the traditional file system is inefficient or not reliable for todays complex & big data. The traditional file system can have security issues or data breaches.

- It ensure data security & consistency by advance data encryption algorithm & use the formate for arranging & retreiving the data easily.
- Also recovery DBMS is capable of getting data recovery by creating replicas, etc. on the other hand it is hard to do in the traditional file system.

Ans-2.



Que-3.
Ans-



Que-4.
Ans= ER Model is important while designing the data because it tells as follows:-

① It help understanding the DB visually.
② It help creating optimize, scalable relationships
③ It help creating integrity constrains & relationship happing.
④ It help understanding the attributes & entities in more efficient way.
⑤ It help scalling & creating efficient DB.
⑥ It prevent Database inconsistency & duplicate
⑦ It also make the DB secure &

also follows principle of least privilage.

(8.) What could happen if relationship is not correctly defined b/w entities :

1) Data Inconsistency : Data can inconstence like if someone delete the user all the product will still exist with no users.

2) Data Redundancy - (Duplicate Data) Data won't be unique the duplicate data takes more space & inconsistency.

3) Security & Accesibility : The data won't be much secure anyone can easily hack or beach the data.

it may cause can't easily use the